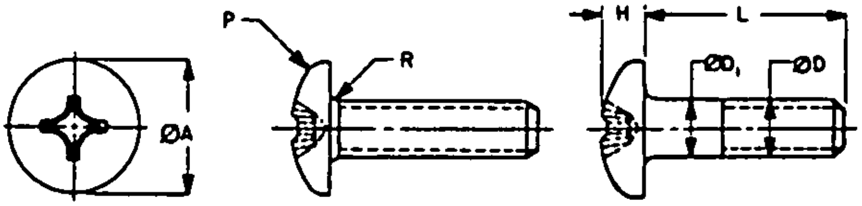
FED. SUP CLASS
5305



USER ACTIVITIES:
ARMY-ME
NAVY-MC,SH

REVIEWER ACTIVITIES:
ARMY-AT,AV,GL,MI
AIR FORCE-85
DLA-IS
NSA-NS

TABLE I. DASH NUMBERS AND DIMENSIONS

ØD NOMINAL SIZE		#2 .0860	#4 .1120	#6 .1380	#8 .1640
THREADS PER INCH		56UNC-2A	40UNC-2A	32UNC-2A	32UNC-2A
ØD₁ BODY	MAX MIN	.0860 .0717	.1120 .0925	.1380 .1141	.1640 .1399
ØA HEAD	MAX MIN	.1670 .1530	.2190 .2050	.2700 .2560	.3220 .3060
H HEAD HEIGHT	MAX MIN	.0620 .0530	.0800 .0700	.0970 .0870	.1150 .1050
P HEAD RADIUS	MIN	.0100	.0100	.0150	.0150
R FILLET RADIUS	MAX	.0130	.0170	.0210	.0250
DRIVER SIZE		1	1	2	2
MINIMUM TENSILE STRENGTH LOAD LBS		300	480	730	1,120
L LENGTH	TOLERANCE	DASH NO	DASH NO	DASH NO	DASH NO
.125		1	11	24	39
.188		2	12	25	40
.250		3	13	26	41
.312		4	14	27	42
.375		5	15	28	43
.438		6	16	29	44
.500		7	17	30	45
.562		141 2/	120 2/	123 2/	126 2/
.625	+0 -.031	8	18	31	46
.750		9	19	32	47
.875		10	20	33	48
1.000		142	21	34	49
1.125			121 2/	124 2/	127 2/
1.250	+0 -.062		22 1/	35	50
1.375			122 2/	125 2/	128 2/
1.500			23	36	51
1.750				37	52
2.000				38	53
2.250	+0 -.094				54
2.500					55
2.750					56
3.000					57

1/ INDICATES MANUFACTURER'S NON-STOCK PRODUCTION ITEMS.

2/ INACTIVE FOR NEW DESIGN AFTER 18 JUL 86

Ⓒ ENTIRE STANDARD REVISED

APPROVED 12 DEC 63   REVISED Ⓒ 18 JUL 86

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

MGC N/A

P. A. ARMY-AR Other Cust NAVY-YD AIR FORCE-99	INTERNATIONAL INTEREST	TITLE SCREW, MACHINE-PAN HEAD, CROSS-RECESSED, CORROSION-RESISTANT STEEL, UNC-2A	MILITARY STANDARD MS51957
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES:		PAGE 1 OF 6

DD FORM 1 OCT 71 672-1 (COORDINATED)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

5305-1558

FED. SUP CLASS
5305

USER ACTIVITIES:
ARMY-ME
NAVY-MC,SH

REVIEWER ACTIVITIES:
ARMY-AT,AV,GL,MI
AIR FORCE-85
DLA-IS
NSA-NS

TABLE I. DASH NUMBERS AND DIMENSIONS - CONTINUED

ØD NOMINAL SIZE		#10 .1900	1/4 .2500	5/16 .3125	3/8 .3750
THREADS PER INCH		24UNC-2A	20UNC-2A	18UNC-2A	16UNC-2A
$ØD_1$ BODY	MAX MIN	.1900 .1586	.2500 .2127	.3125 .2712	.3750 .3287
ØA HEAD	MAX MIN	.3730 .3570	.4920 .4730	.6150 .5940	.7400 .7160
H HEAD HEIGHT	MAX MIN	.1330 .1220	.1750 .1620	.2180 .2030	.2610 .2440
P HEAD RADIUS	MIN	.0200	.0350	.0400	.0400
R FILLET RADIUS	MAX	.0290	.0380	.0470	.0560
DRIVER SIZE		2	3	4	4
MINIMUM TENSILE STRENGTH LOAD LBS		1,400	2,540	4,190	6,200
L LENGTH	TOLERANCE	DASH NO	DASH NO	DASH NO	DASH NO
.188	+0 -.031	58[1]			
.250		59			
.312		60	76		
.375		61	77	92	
.438		62	78	93	
.500		63	79	94	107
.562		129[2]	132[2]	135[2]	138[2]
.625		64	80	95	108
.750		65	81	96	109
.875		66	82	97	110
1.000		67	83	98	111
1.125	+0 -.062	130[2]	133[2]	136[2]	139[2]
1.250		68	84	99	112
1.375		131[2]	134[2]	137[2]	140[2]
1.500		69	85	100	113
1.750		70	86	101	114
2.000		71	87	102	115
2.250	+0 -.094	72	88	103	116
2.500		73	89	104	117
2.750		74[1]	90[1]	105[1]	118
3.000		75[1]	91[1]	106[1]	119

[1] INDICATES MANUFACTURER'S NON-STOCK PRODUCTION ITEMS.

[2] INACTIVE FOR NEW DESIGN AFTER

APPROVED 12 DEC 63 REVISED Ⓒ FOR CHANGES SEE PAGES 1,2,3,4,5 AND 6

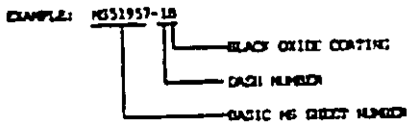
This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A. ARMY-AR Other Cust NAVY-YD AIR FORCE-99	INTERNATIONAL INTEREST	TITLE SCREW, MACHINE-PAN HEAD, CROSS-RECESSED, CORROSION-RESISTANT STEEL,UNC-2A	MILITARY STANDARD MS51957
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES:		PAGE 2 OF 6

FED SUP CLASS
5305

REQUIREMENTS:

1. MATERIAL: CORROSION RESISTANT STEEL, 300 SERIES, IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.
2. CLEANED, DESCALED AND PASSIVATED: SCREWS SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH ASTM A380.
3. PROTECTIVE COATING AND SURFACE TREATMENT: WHEN SPECIFIED SCREWS SHALL BE BLACK OXIDE COATED IN ACCORDANCE WITH MIL-C-13924, CLASS 4. (WHEN BLACK OXIDE COATING IS REQUIRED, THE DASH NUMBER SHALL BE FOLLOWED BY A "B").
4. MAGNETIC PERMEABILITY: SCREWS SHALL MEET THE MAGNETIC PERMEABILITY REQUIREMENTS OF THE PROCUREMENT SPECIFICATION.
5. THREADS: THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/2. ACCEPTABILITY OF SCREW THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 21. THREADS SHALL EXTEND TO WITHIN 2 THREADS OR LESS OF THE BEARING SURFACE, EXCEPT THAT ALL LENGTHS OVER 2 INCHES SHALL HAVE A MINIMUM COMPLETE THREAD LENGTH OF 1 3/4 INCHES.
6. RECESS: THE RECESS SHALL BE IN ACCORDANCE WITH MS9006.
7. TENSILE STRENGTH: THE MINIMUM TENSILE STRENGTH IN LOAD POUNDS, INDICATED FOR EACH SIZE IN TABLE I, IS BASED ON 80,000 psi MINIMUM TENSILE STRENGTH. LOAD POUNDS ARE CALCULATED BY THE STRESS AREAS INDICATED IN FED-STD-H28/2.
8. PART NUMBER: THE PART NUMBER SHALL CONSIST OF THE BASIC MS SHEET NUMBER, FOLLOWED BY A DASH NUMBER FROM TABLE I AND A CODE LETTER "B" FOR BLACK OXIDE, IF APPLICABLE.



MS51957-1B INDICATES - SCREW, MACHINE - PAN HEAD, CROSS-RECESSED, CORROSION RESISTANT STEEL; BLACK OXIDE COATING; .0860-56UNC-2A NOMINAL THREAD SIZE; .125 THREAD LENGTH.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
3. REFERENCED GOVERNMENT (OR NONGOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DoDISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

USER ACTIVITIES:
ARMY-ME
NAVY-MC,SH

REVIEWER ACTIVITIES:
ARMY-AT,AV,GL,MI
AIR FORCE-85
DLA-IS
NSA-NS

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

APPROVED 12 DEC 63 REVISED Ⓒ FOR CHANGES SEE PAGES 1,2,3,4,5 AND 6

P. A. ARMY-AR Other Cust NAVY-YD AIR FORCE-99	INTERNATIONAL INTEREST	TITLE SCREW, MACHINE-PAN HEAD, CROSS-RECESSED, CORROSION-RESISTANT STEEL, UNC-2A	MILITARY STANDARD MS 51957
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES:		PAGE 3 OF 6

FED. SUP CLASS
5305

THE SCREWS COVERED BY THE DASH NUMBERS GIVEN IN MS16200, MS35216, MS35233, AN515 IN PART AND AN526 IN PART ARE CANCELLED/INACTIVATED AFTER THE DATES INDICATED ON THE DOCUMENTS. USE THE DASH NUMBERS GIVEN IN THE PRECEDING SHEETS. THE CANCELLED SCREWS CANNOT ALWAYS REPLACE THE NEW SCREWS AND SHOULD BE USED UNTIL EXISTING STOCKS ARE DEPLETED. USE ONLY THE NEW SCREWS FOR DESIGN AND REPLACEMENT. REPLACEMENT SHALL BE IN ACCORDANCE WITH THIS TABLE.

TABLE II. INTERCHANGEABILITY

CROSS REFERENCE OF PART NUMBERS						
CANCELLED						NEW
MS16200 MS35233	MS35216	AN515		AN526		MS51957
DASH NUMBER						DASH NUMBER
1	1	C2R2	C2-2			1
2	2	C2R3	C2-3			2
3	3	C2R4	C2-4			3
4	4	C2R5	C2-5			4
5	5	C2R6	C2-6			5
6	6	C2R7	C2-7			6
7	7	C2R8	C2-8			7
8	8	C2R10	C2-10			8
9	9	C2R12	C2-12			9
10						10
		C3R2 thru C3R16	C3-2 thru C3-16			2/
11	10					11
12	11	C4R3	C4-3			12
13	12	C4R4	C4-4			13
14	13	C4R5	C4-5			14
15	14	C4R6	C4-6			15
16	15	C4R7	C4-7			16
17	16	C4R8	C4-8			17
18	17	C4R10	C4-10			18
19	18	C4R12	C4-12			19
20	19	C4R14	C4-14			20
21	20	C4R16	C4-16			21
		C4R18	C4-18			121 1/
22		C4R20	C4-20			22
		C4R22	C4-22			122 1/
23		C4R24	C4-24			23
		C4R26	C4-26			2/
		C4R28	C4-28			2/
		C5R3 thru C5R28	C5-3 thru C5R28			2/
24	21					24
25	22	C6R3	C6-3			25
26	23	C6R4	C6-4	C632R4	C632-4	26
27	24	C6R5	C6-5	C632R5	C632-5	27
28	25	C6R6	C6-6	C632R6	C632-6	28
29	26	C6R7	C6-7	C632R7	C632-7	29
30	27	C6R8	C6-8	C632R8	C632-8	30
				C632R9	C632-9	123 1/
31	28	C6R10	C6-10	C632R10	C632-10	31
32	29	C6R12	C6-12	C632R12	C632-12	32
33	30	C6R14	C6-14	C632R14	C632-14	33
34	31	C6R16	C6-16	C632R16	C632-16	34
		C6R18	C6-18	C632R18	C632-18	124 1/
35	32	C6R20	C6-20	C632R20	C632-20	35
		C6R22	C6-22	C632R22	C632-22	125 1/
36	33	C6R24	C6-24	C632R24	C632-24	36
		C6R26	C6-26	C632R26	C632-26	2/
37	34	C6R28	C6-28	C632R28	C632-28	37
		C6R30	C6-30	C632R30	C632-30	2/
38	35	C6R32	C6-32	C632R32	C632-32	38
		C6R36	C6-36	C632R36	C632-36	2/
		C6R40	C6-40	C632R40	C632-40	2/
				C632R44	C632-44	2/
				C632R48	C632-48	2/

1/ NEW DASH NUMBERS ADDED 26 AUG 1968

2/ NO REPLACEMENT

USER ACTIVITIES:
ARMY-ME
NAVY-MC,SH

REVIEWER ACTIVITIES:
ARMY-AT,AV,GL,MI
AIR FORCE-85
DLA-IS
NSA-NS

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

APPROVED 12 DEC 63 REVISED © FOR CHANGES SEE PAGES 1,2,3,4,5 AND 6

P. A ARMY-AR Other Cust NAVY-YD AIR FORCE-99	INTERNATIONAL INTEREST	TITLE SCREW, MACHINE-PAN HEAD, CROSS-RECESSED, CORROSION-RESISTANT STEEL, UNC-2A	MILITARY STANDARD MS51957
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES:		PAGE 4 OF 6

USED ACTIVITIES:
ARMY-ME
NAVY-MC, SH

REVIEWER ACTIVITIES:
ARMY-AT,AV,GL,MI
AIR FORCE-85
DLA-IS
NSA-NS

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

TABLE II. INTERCHANGEABILITY - CONTINUED

CROSS REFERENCE OF PART NUMBERS						
CANCELLED						NEW
MS16200 MS35233	MS35216	AN515		AN526		MS51957
DASH NUMBER						DASH NUMBER
39	36					39
40	37					40
41	38	C8R4	C8-4	C832R4	C832-4	41
42	39	C8R5	C8-5	C832R5	C832-5	42
43	40	C8R6	C8-6	C832R6	C832-6	43
44	41	C8R7	C8-7	C832R7	C832-7	44
45	42	C8R8	C8-8	C832R8	C832-8	45
				C832R9	C832-9	126[1]
46	43	C8R10	C8-10	C832R10	C832-10	46
47	44	C8R12	C8-12	C832R12	C832-12	47
48	45	C8R14	C8-14	C832R14	C832-14	48
49	46	C8R16	C8-16	C832R16	C832-16	49
		C8R18	C8-18	C832R18	C832-18	127[1]
50	47	C8R20	C8-20	C832R20	C832-20	50
		C8R22	C8-22	C832R22	C832-22	128[1]
51	48	C8R24	C8-24	C832R24	C832-24	51
		C8R26	C8-26	C832R26	C832-26	[2]
52	49	C8R28	C8-28	C832R28	C832-28	52
		C8R30	C8-30	C832R30	C832-30	[2]
53	50	C8R32	C8-32	C832R32	C832-32	53
54		C8R36	C8-36	C832R36	C832-36	54
55		C8R40	C8-40	C832R40	C832-40	55
56		C8R44	C8-44	C832R44	C832-44	56
57		C8R48	C8-48	C832R48	C832-48	57
58						58
59	51	C10R4	C10-4	C1024R4	C1024-4	59
60	52	C10R5	C10-5	C1024R5	C1024-5	60
61	53	C10R6	C10-6	C1024R6	C1024-6	61
62	54	C10R7	C10-7	C1024R7	C1024-7	62
63	55	C10R8	C10-8	C1024R8	C1024-8	63
				C1024R9	C1024-9	129[1]
64	56	C10R10	C10-10	C1024R10	C1024-10	64
65	57	C10R12	C10-12	C1024R12	C1024-12	65
66	58	C10R14	C10-14	C1024R14	C1024-14	66
67	59	C10R16	C10-16	C1024R16	C1024-16	67
		C10R18	C10-18	C1024R18	C1024-18	130[1]
68	60	C10R20	C10-20	C1024R20	C1024-20	68
		C10R22	C10-22	C1024R22	C1024-22	131[1]
69	61	C10R24	C10-24	C1024R24	C1024-24	69
		C10R26	C10-26	C1024R26	C1024-26	[2]
70	62	C10R28	C10-28	C1024R28	C1024-28	70
		C10R30	C10-30	C1024R30	C1024-30	[2]
71	63	C10R32	C10-32	C1024R32	C1024-32	71
72	64	C10R36	C10-36	C1024R36	C1024-36	72
73	65	C10R40	C10-40	C1024R40	C1024-40	73
74		C10R44	C10-44	C1024R44	C1024-44	74
75		C10R48	C10-48	C1024R48	C1024-48	75
76	66	C416R5	C416-5			76
77	67	C416R6	C416-6	C420R6	C420-6	77
78	68	C416R7	C416-7	C420R7	C420-7	78
79	69	C416R8	C416-8	C420R8	C420-8	79
				C420R9	C420-9	132[1]
80	70	C416R10	C416-10	C420R10	C420-10	80
81	71	C416R12	C416-12	C420R12	C420-12	81
82	72	C416R14	C416-14	C420R14	C420-14	82
83	73	C416R16	C416-16	C420R16	C420-16	83
		C416R18	C416-18	C420R18	C420-18	133[1]

[1] NEW DASH NUMBERS ADDED 26 AUG 1968.

[2] NO REPLACEMENT

APPROVED 12 DEC 63 REVISED Ⓒ FOR CHANGES SEE PAGES 1,2,3,4,5 AND 6

P. A. ARMY-AR Other Cust NAVY-YD AIR FORCE-99	INTERNATIONAL INTEREST	TITLE SCREW, MACHINE-PAN HEAD, CROSS-RECESSED, CORROSION-RESISTANT STEEL, UNC-2A	MILITARY STANDARD MS 51957
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES		PAGE 5 OF 6

FED. SUP CLASS
5305

TABLE 11. INTERCHANGEABILITY - CONTINUED

CROSS REFERENCE OF PART NUMBERS						
CANCELLED						NEW
MS16200 MS35233	MS35216	AN515		AN526		MS51957
DASH NUMBER						DASH NUMBER
84	74	C416R20	C416-20	C420R20	C420-20	84
		C416R22	C416-22	C420R22	C420-22	134 1/
85	75	C416R24	C416-24	C420R24	C420-24	85
		C416R26	C416-26	C420R26	C420-26	2/
86	76	C416R28	C416-28	C420R28	C420-28	86
		C416R30	C416-30	C420R30	C420-30	2/
87	77	C416R32	C416-32	C420R32	C420-32	87
88	78	C416R36	C416-36	C420R36	C420-36	88
89	79	C416R40	C416-40	C420R40	C420-40	89
90		C416R44	C416-44	C420R44	C420-44	90
91		C416R48	C416-48	C420R48	C420-48	91
92	80					92
93	81					93
94	82	C516R8	C516-8			94
95	83	C516R10	C516-10			95
96	84	C516R12	C516-12			96
97	85	C516R14	C516-14			97
98	86	C516R16	C516-16			98
		C516R18	C516-18			136 1/
99	87	C516R20	C516-20			99
		C516R22	C516-22			137 1/
100	88	C516R24	C516-24			100
		C516R26	C516-26			2/
101	89	C516R28	C516-28			101
		C516R30	C516-30			2/
102	90	C516R32	C516-32			102
103	91	C516R36	C516-36			103
104	92	C516R40	C516-40			104
105		C516R44	C516-44			105
106		C516R48	C516-48			106
107	93	C616R8	C616-8			107
108	94	C616R10	C616-10			108
109	95	C616R12	C616-12			109
110	96	C616R14	C616-14			110
111	97	C616R16	C616-16			111
		C616R18	C616-18			139 1/
112	98	C616R20	C616-20			112
		C616R22	C616-22			140 1/
113	99	C616R24	C616-24			113
		C616R26	C616-26			2/
114	100	C616R28	C616-28			114
		C616R30	C616-30			2/
115	101	C616R32	C616-32			115
116	102	C616R36	C616-36			116
117	103	C616R40	C616-40			117
118	104	C616R44	C616-44			118
119	105	C616R48	C616-48			119

1/ NEW DASH NUMBERS ADDED 26 AUG 1968

2/ NO REPLACEMENT

USER ACTIVITIES:
ARMY-ME
NAVY-MC,SH

REVIEWER ACTIVITIES:
ARMY-AT,AV,GL,MI
AIR FORCE-85
DLA-IS
NSA-NS

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

APPROVED 12 DEC 63 REVISED © FOR CHANGES SEE PAGES 1,2,3,4,5 AND 6

P. A. ARMY-AR Other Cust NAVY-YD AIR FORCE-99	INTERNATIONAL INTEREST	TITLE SCREW, MACHINE-PAN HEAD, CROSS-RECESSED, CORROSION-RESISTANT STEEL, UNC-2A	MILITARY STANDARD MS51957
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES:		PAGE 6 OF 6